2026 MIDDLE SCHOOL STEM CAMPS

# From Trees to Tech

## INSTRUCTOR GUIDE PACKET

Facilitation guides for every primary and backup activity.

IN THIS PACKET

INSTRUCTOR · TTT-01 · BIOLOGY + ENERGY

# MudWatt Bioelectric League

Turn a cup of mud into a living battery

**At a glance:** Can your team make mud generate electricity and defend your design with data? About 80 min; 4 to 6 teams.

## MATERIALS

- carbon felt electrodes
- 24 oz deli cups
- 100 kΩ resistors
- alligator leads
- multimeters, one per team
- sugar cubes for fuel
- fresh soil or mud samples
- nitrile gloves
- labels and masking tape
- handwashing supplies

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Which design change moved your readings the most, and how do you know?
- Why must the anode stay buried and the cathode stay in air?
- How is your weekly log like the work of an energy engineer?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS |
|---|---|
| Peak voltage (mV) | 35 |
| Daily data log quality | 20 |
| Controlled-variable design | 15 |
| Redesign or maintenance plan | 15 |
| Final explanation | 15 |
| **Total** | **100** |

**SOURCES** Ohio State Microbial Fuel Cell Learning Center

# Forest Sensor Sprint

Site the smartest place for a campus tree sensor

**At a glance:** Find the smartest place to install a future campus tree sensor. About 80 min; 4 to 6 teams.

## MATERIALS

- 4 clipboards (one per team; 6 on hand)
- 4 paper clinometers (one per team; print 6 to 8 with spares)
- 4 temperature and humidity meters (one per team; 12 on hand)
- 4 soil meters (one per team; 6 on hand). The same meter also reads light, so no separate light meter is needed.
- 4 route cards (one per team; print 6 to 8 with spares)
- pencils

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Which measurement most influenced your choice?
- Where did two nearby spots disagree, and why?
- What would change your recommendation in winter?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS |
|---|---|
| Data accuracy | 30 |
| Route efficiency | 20 |
| Site recommendation evidence | 30 |
| Teamwork and field conduct | 20 |
| **Total** | **100** |

**SOURCES** GLOBE Observer Trees; NASA MyNASAData Urban Heat Island

# Seed Dispersal Derby

Engineer a seed that escapes its parent tree

**At a glance:** Build a seed that escapes the parent tree without engines or throwing. About 80 min; 4 to 6 teams.

## MATERIALS

- paper
- cardstock
- paper clips
- masking tape
- string
- fan or marked test lane
- timers

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Which change improved hang time without losing distance?
- Why does spinning slow the fall?
- Which tree's strategy did your final seed copy?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS |
| --- | --- |
| Distance traveled | 35 |
| Hang time | 25 |
| Target landing | 15 |
| Biomimicry explanation | 15 |
| Redesign improvement | 10 |
| **Total** | **100** |

**SOURCES** Science Buddies Seed Dispersal; Project Learning Tree

# Xylem Pipeline Relay

Move water uphill the way a tree does

**At a glance:** Move water uphill like a tree, faster and cleaner than rival teams. About 80 min; 4 to 6 teams.

## MATERIALS

- paper towels
- cotton string
- felt strips
- cups
- food coloring
- trays
- rulers
- elevation blocks

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Which material moved water best, and why?
- How did slope change your delivery?
- Where does a real tree use this same effect?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS |
|---|---|
| Water delivered | 40 |
| Speed | 20 |
| Least spill | 15 |
| Design explanation | 15 |
| Redesign gain | 10 |
| **Total** | **100** |

**SOURCES** Science Buddies; plant water transport references

# Greenhouse Climate Controller

Match plants to the right climate zone

**At a glance:** Run the greenhouse controls for a plant that cannot complain, only wilt. About 80 min; 4 to 6 teams.

## MATERIALS

- plant profile cards
- control dial boards
- clipboards
- dry erase markers
- tour clue sheets

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Which plant was hardest to place, and why?
- What tradeoff did you have to accept?
- How do real growers handle conflicting needs?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS |
|---|---|
| Correct setting choices | 30 |
| Use of tour evidence | 25 |
| Explanation | 20 |
| Layout efficiency | 15 |
| Teamwork | 10 |
| **Total** | **100** |

**SOURCES** Temple Ambler Greenhouse

INSTRUCTOR · TTT-06 · BIOMIMICRY

# Pinecone Weather Machine

A humidity sensor with no battery and no code

**At a glance:** Build a weather sensor with no battery and no code. About 80 min; 4 to 6 teams.

## MATERIALS

- dry pine cones
- index cards
- wood skewers
- spray bottles
- rulers
- masking tape
- paper fasteners

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Why does a two-layer strip bend when only one layer swells?
- What made your indicator faster?
- Where would a no-power humidity sensor be useful?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS |
|---|---|
| Response accuracy | 35 |
| Response speed | 20 |
| Readable output scale | 20 |
| Biomimicry explanation | 15 |
| Redesign quality | 10 |
| **Total** | **100** |

**SOURCES** AskNature Pine Cones Strategy

# Pollinator Network Draft and Build

Design a habitat that works all season

**At a glance:** Draft a pollinator habitat that works all season, not just on the prettiest day. About 80 min; 4 to 6 teams.

## MATERIALS

- plant cards
- pollinator cards
- grid boards
- stickers
- markers
- season tokens

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Where was the biggest gap in your bloom calendar?
- Why does clumping help pollinators?
- What happens to the network if one season is missing?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS |
| --- | --- |
| Seasonal coverage | 30 |
| Pollinator fit | 25 |
| Native and clumping logic | 20 |
| Layout clarity | 15 |
| Defense | 10 |
| **Total** | **100** |

**SOURCES** USDA Pollinators; US Forest Service pollinator guidance

INSTRUCTOR · TTT-08 · FIELD SCIENCE

# Arboretum Eco-Quest

Solve outdoor ecology checkpoints

**At a glance:** Read the living collection and solve the checkpoints before time runs out. About 80 min; 4 to 6 teams.

## MATERIALS

- clue cards
- route map
- clipboards
- pencils
- evidence token bags
- optional QR codes

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Which clue was most reliable for telling trees apart?
- Where did a key send you wrong, and why?
- What makes a good piece of field evidence?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS |
|---|---|
| Checkpoint accuracy | 50 |
| Evidence quality | 20 |
| Pace | 20 |
| Teamwork and conduct | 10 |
| **Total** | **100** |

**SOURCES** Temple Ambler Arboretum; Arboretum Explorer

# Minecraft Tree World Resilience Cup

Design a climate-resilient landscape

**At a glance:** Design a landscape that survives stress because of how it is built, not luck. About 80 min; 4 to 6 teams.

## MATERIALS

- Minecraft Education devices and template or paper grid kit
- rubric cards
- display timer

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Which feature would help most in a real storm?
- Why does diversity add resilience?
- What in your build would fail first, and how would you fix it?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS |
|---|---|
| Resilience features | 30 |
| Biodiversity support | 25 |
| Realism | 20 |
| Explanation | 15 |
| Build polish | 10 |
| **Total** | **100** |

**SOURCES** Minecraft Education Biodiversity

# Tree Ring Climate Detective

Read rings to reconstruct past climate

**At a glance:** Read the rings to reconstruct the climate events a tree lived through. About 80 min; 4 to 6 teams.

## MATERIALS

- printed ring cards
- wood cookies or images
- answer boards
- timers
- claim-evidence cards

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Which pattern was easiest to misread?
- How is a tree ring like and unlike a thermometer?
- What other natural records store past climate?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS |
|---|---|
| Correct inferences | 40 |
| Evidence justification | 20 |
| Speed | 20 |
| Final synthesis | 20 |
| **Total** | **100** |

**SOURCES** NOAA Tree Rings; EPA Tree Rings, Living Records of Climate

# Lotus Leaf Surface Sprint

Engineer a self-cleaning, water-shedding surface

**At a glance:** Build a surface that sheds water and dirt the way a lotus leaf does. About 80 min; 4 to 6 teams.

## MATERIALS

- wax paper
- cardstock
- sandpaper strips
- masking tape
- pipettes
- water
- pepper or cocoa residue
- ramps

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Why does a rough, waxy surface shed water better than a smooth one?
- What carried the dirt away?
- Where would a self-cleaning surface be useful?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS |
|---|---|
| Fast clean runoff | 35 |
| Least residue | 25 |
| Surface design explanation | 20 |
| Redesign gain | 10 |
| Craftsmanship | 10 |
| **Total** | **100** |

**SOURCES** AskNature Lotus Leaf Self-Cleaning Surface

# Leaf Stomata Microscope Detective

Count stomata and rank leaves by water strategy

**At a glance:** Count the breathing pores and rank leaves by how they manage water. About 80 min; 4 to 6 teams.

## MATERIALS

- leaf samples
- clear tape
- clear nail polish or prepared slides
- USB microscopes or magnifiers
- grid sheets
- gloves

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Why count several fields instead of one?
- What does a high stomata count suggest about a leaf's habitat?
- Where might error sneak into your count?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS |
|---|---|
| Slide or peel quality | 25 |
| Counting accuracy | 30 |
| Ranking evidence | 25 |
| Team data table | 10 |
| Cleanup and care | 10 |
| **Total** | **100** |

**SOURCES** Leaf stomata microscopy adaptations; Science Buddies plant biology

BACKUP ACTIVITIES

# Ready alternates

# Root Grip Erosion Rescue

Can roots hold a slope together in a storm?

**At a glance:** Can roots hold a slope together during a storm? About 60 min; 4 to 6 teams.

## MATERIALS

- Trays
- Soil
- String or yarn roots
- Craft sticks
- Watering cups

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Why did the rooted slope hold more soil?
- How would deeper roots change the result?
- Where is this used to protect real hillsides?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS |
|---|---|
| Soil retained | 40 |
| Runoff control | 25 |
| Design explanation | 20 |
| Cleanup | 15 |
| **Total** | **100** |

**SOURCES** EPA green infrastructure and erosion lessons

INSTRUCTOR · TTB-02 · FIELD SCIENCE

# Tree Height Triangulation Shootout

Measure a tree without climbing it

**At a glance:** Measure a tree without climbing it. About 60 min; 4 to 6 teams.

## MATERIALS

- Paper clinometers
- String and washers
- Rulers and tape measures
- Clipboards

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- What was your biggest source of error?
- Why does eye height matter in the calculation?
- Where is remote height measurement used in the field?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS |
|---|---|
| Accuracy | 50 |
| Method clarity | 20 |
| Speed | 20 |
| Teamwork | 10 |
| **Total** | **100** |

**SOURCES** GLOBE Trees Activities

# Urban Heat Shade Dash

Find the coolest route through a hot campus

**At a glance:** Find the coolest route through a hot campus. About 60 min; 4 to 6 teams.

## MATERIALS

- Thermometers or IR thermometers
- Campus maps
- Clipboards

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- How big was the sun-to-shade temperature gap?
- What surface was hottest, and why?
- How do cities use shade to cool streets?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS |
|---|---|
| Cool route design | 35 |
| Data quality | 25 |
| Reasoning | 20 |
| Communication | 20 |
| **Total** | **100** |

**SOURCES** NASA MyNASAData Urban Heat Island

# Photosynthesis Float-Off Playoffs

Make leaf disks float by making oxygen

**At a glance:** Make leaf disks float by producing oxygen. About 60 min; 4 to 6 teams.

## MATERIALS

- Spinach leaves
- Baking soda solution
- Cups
- Oral syringes
- no needle
- Timers

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Why does oxygen make the disks float?
- Which variable changed the rate most?
- How is this evidence of photosynthesis?

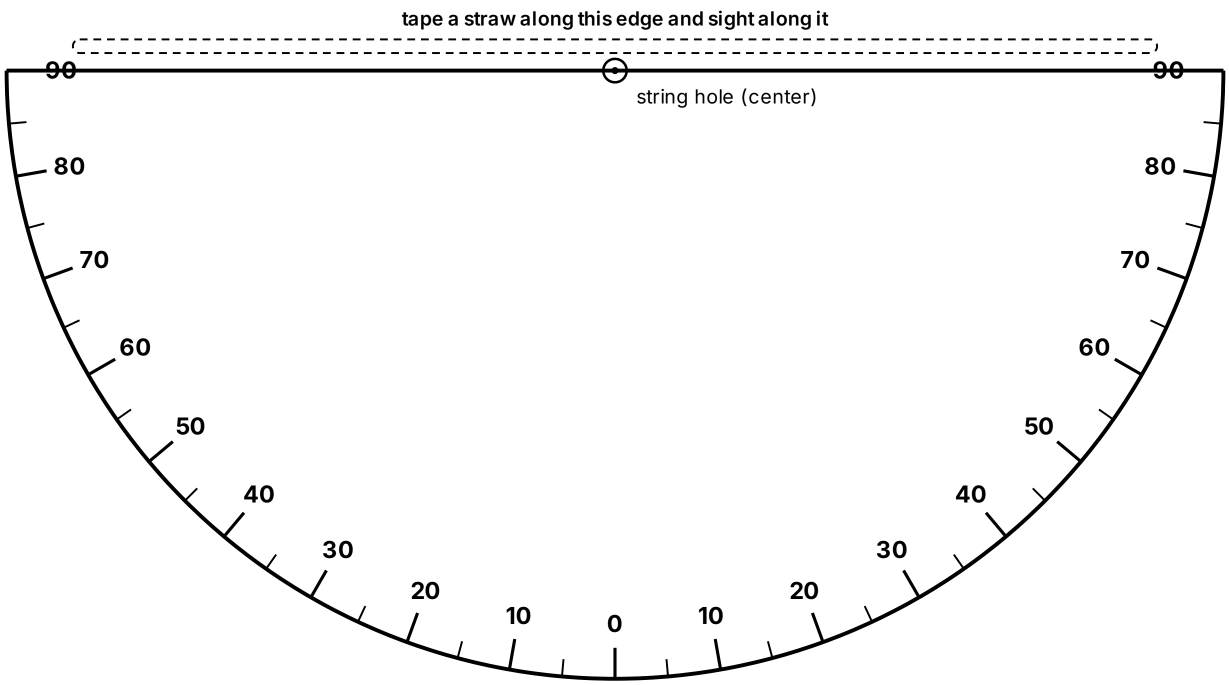
## SCORING (100 POINTS)

| SCORING CRITERION | POINTS |
|---|---|
| Performance | 35 |
| Controlled-variable design | 25 |
| Data table | 20 |
| Explanation | 10 |
| Cleanup | 10 |
| **Total** | **100** |

**SOURCES** Exploratorium Photosynthetic Floatation

Print on cardstock for TTT-02 Forest Sensor Sprint. Clinometer: one per page, print 6 to 8. Route cards: two per page, print 3 to 4 sheets for 6 to 8 cards.

## Paper clinometer




Cut along the flat top edge and the curved outline.

### ASSEMBLE

01 Cut out the half circle along the curved outline.

02 Tape a straight straw along the top straight edge.

03 Punch the center dot, thread a string through it, and tie a washer or paper clip on the end as a weight.

04 Sight the treetop through the straw, let the string hang free, then pinch it against the scale and read the number.

05 Self-check: sight something level (reads 0) and straight up (reads 90).

### FIND THE TREE HEIGHT

Tree height = eye height + ( distance to the tree × tan of the angle ). Use the standoff distance the instructor marked.

| ANGLE | TAN |
|---|---|
| 10 | 0.18 |
| 20 | 0.36 |
| 30 | 0.58 |
| 40 | 0.84 |
| 45 | 1.00 |
| 50 | 1.19 |
| 60 | 1.73 |
| 70 | 2.75 |

Example: eye height 1.5 m, distance 10 m, angle 40° gives 1.5 + 10 × 0.84 = about 9.9 m.

## Team route card

### From Trees to Tech · TTT-02 Forest Sensor Sprint · Team Route Card

Team ________________________   Date ______________

Plan an order that avoids backtracking, take a reading at each checkpoint, then recommend the best sensor site using your numbers.

| CHECKPOINT | VISIT ORDER | TEMP (°F) | HUMIDITY (%) | LIGHT (LUX) | SOIL MOISTURE (1 TO 10) | NOTES |
|---|---|---|---|---|---|---|
| Reference (calibrate) | | | | | | |
| A | | | | | | |
| B | | | | | | |
| C | | | | | | |
| D | | | | | | |
| E | | | | | | |

Our recommended sensor site: ______________________________________

Why (use your data): ____________________________________________

The checkpoints are the spots the instructor marked outside.

cut here


### From Trees to Tech · TTT-02 Forest Sensor Sprint · Team Route Card

Team ________________________   Date ______________

Plan an order that avoids backtracking, take a reading at each checkpoint, then recommend the best sensor site using your numbers.

| CHECKPOINT | VISIT ORDER | TEMP (°F) | HUMIDITY (%) | LIGHT (LUX) | SOIL MOISTURE (1 TO 10) | NOTES |
|---|---|---|---|---|---|---|
| Reference (calibrate) | | | | | | |
| A | | | | | | |
| B | | | | | | |
| C | | | | | | |
| D | | | | | | |
| E | | | | | | |

Our recommended sensor site: ______________________________________

Why (use your data): ____________________________________________

The checkpoints are the spots the instructor marked outside.